EXERCISE 1

Given $f(x) = \sqrt{1-x}$ and $g(x) = x^2$, find :

1. $(f \circ g)(x)$ [3 marks]

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2. the domain and range of $(f \circ g)(x)$ [4 marks]

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EXERCISE 2 Given $f(x) = 2 + x$ and $g(x) = 3 - x$, find :

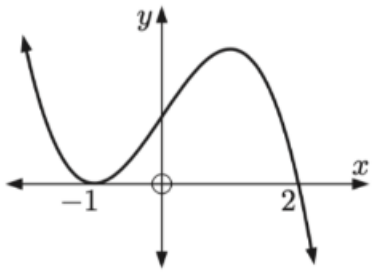
1. $(f \circ f)(-5)$ [3 marks]

2. $(g \circ f)(3)$ [3 marks]

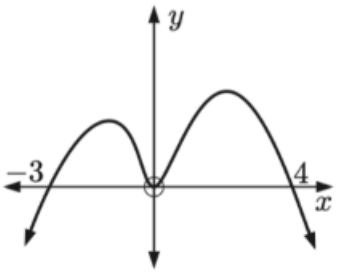
EXERCISE 3 [10 marks]

Draw sign diagrams for these graphs :

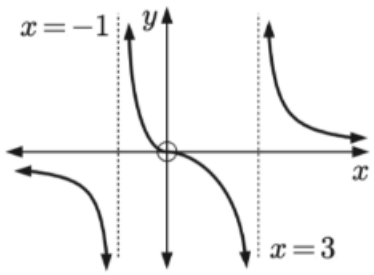
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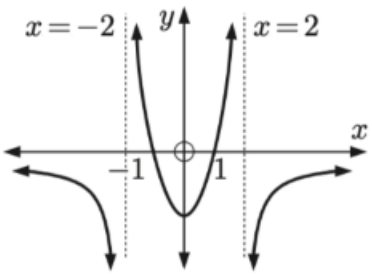
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EXERCISE 4 [6 marks]

Draw sign diagrams for :

1. $(5-x)(1-2x)$

2. $2-8x^2$

3. $-2x^2+9x+5$